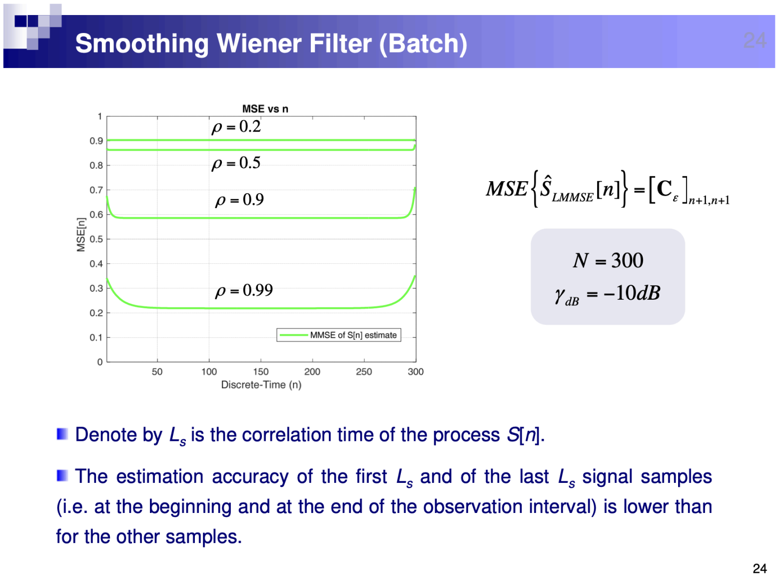
DISCLAIMER: If you don't like the total vote, you probably have to do another time, also the written part.

FIRST QUESTION:

1. Asked about the large number law and how it is connected to the sample estimator (all the demonstration,from the Binary random variable to the frequency) . Also what's the condition for the estimator to be consistent (var<+inf)(Third pdf from 23-32, 41-48 and 56-65).
2. Talk about the Least Squares estimator, from the general definition (slide 4 chapter 5) to the geometric interpretation in the scalar case (slides 10-23 chapter 5). Explain the differences between LS and ML. Apply the least squares method to the estimation of the amplitude of a signal with known shape (example from slides 47–48, Chapter 5).
3. Linear prediction
4. Describe an ARMA model and how to derive its coefficient (do all the demonstration and write all what you remember)
5. LMMSE estimator: its definition and geometrical interpretation, the derivation of the formula in the scalar case (regression line), and the writing of the formula in the vectorial case. In general pros and cons and when it is used.
6. [written one] Talk about LMMSE, focus on differences with MMSE, geometrical interpretation, when it coincides with MMSE
7. ARMA models all what you remember, in particular why they are useful for the frequency domain.
8. Everything you know about MAP estimation and relation with the MMSE and ML estimators

SECOND QUESTION

1. Talk about this graph:

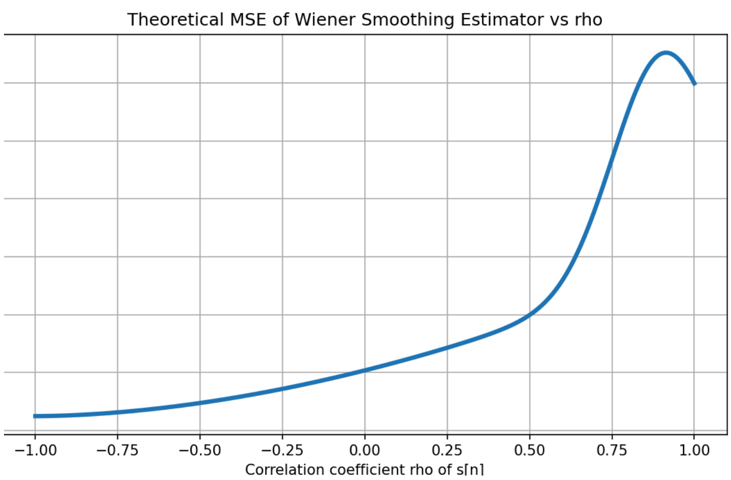


2. In the context of the Wiener smoothing filter, suppose we observe N samples of a process x[n] = s[n] + w[n], with s and w uncorrelated.

   s is an AR(1) process with correlation coefficient rho;

   w is an AR(1) process with rho equal to 0.9. Therefore, we know that it has a narrowband PSD that simulates a low-pass behavior.

   We want to estimate s[N/2] using the Wiener smoothing filter. How will the plot of the MSE look with respect to the correlation coefficient rho of the process s[n]?



One must reason based on the fact that the Wiener smoothing filter works better when the spectra of the process s and the noise w are different. Therefore, the MSE will have a minimum for rho equal to minus 0.9, that is when the process has only high-frequency components which contrast with the low-frequency components of the noise. On the other hand, the MSE will reach a maximum when the rho of the process s coincides with that of the process w. In this case the two signals are similar and the Wiener filter cannot estimate well, since both noise and signal have only low-frequency components.

3. How do we estimate the frequency F0 in a sinusoidal/cosinusoidal process when it is not known: he just wanted to know that the peak frequency is found as the peak of the periodogram (ML estimator). Then, he asked to comment on the catastrophic errors that affect the estimate and to explain the MSE/CRB vs SNR graph of the estimate. He asked me which portion of the graph was due to the catastrophic errors and why it had a linear trend, such that the MSE increased linearly as the SNR decreased (he wanted to know that it is a sort of vicious cycle: as the SNR becomes worse, increasing the rate of catastrophic errors, that, by increasing, further worsen the SNR, and so on)

4. Describe in words the estimation of Phase amplitude and frequency with ML (all passages but in THIS CASE, didn't ask the demonstration, just to explain how by words), describe the graph of page 217 of chapter 4 (he opened the graph to let you comment it)

5. Describe the example and explain how we can approximate an ARMA model with an AR of higher order (the example was one in the ARMA PP and he asked to explain the difference among various Graphs). Explain why periodogram is not a consistent estimate in case of the continuous part of pdf

6. Describe how Wiener smoothing filter works (he want to know if is LTI or LTV and how change in relationship with rho in case of AR(1)).
7. Why does the MAP estimator tend to be similar to the ML estimator when N->inf. The reason was because when N increases, since the ML is a consistent estimator, the variance decreases and the pdf tends to be similar to a dirac delta. In this way, the product with the a priori pdf tends to still be a dirac delta so the ML estimator prevails
8. Draw a graph of MSE in relation to the rho of the signal when W is AWGN. You need to put a maximum in 0 and then it looks kind of like a gaussian